TEXORA · LECTURE SHEET

Color and Weave Effects

Give the warp and the weft each a repeating order of coloured threads, and the weave itself decides which colour shows at every intersection. Out of that single mechanism come hairlines, checks, shepherd's checks and the dogtooth of suiting — patterns woven with no printing, no jacquard, and no extra yarn. Every effect in this sheet is computed cell by cell from the weave and the two colour orders.

TE-3205 · FABRIC STRUCTURE AND DESIGN · TOPIC 6 · LEVEL 3 TERM 2

1. The one rule

At any intersection exactly one thread is on the face: the **warp end, where the design square is marked** (end up), or the **weft pick, where the square is blank**. Therefore:

visible colour at (end, pick) = warp colour of that end if the square is marked, else weft colour of that pick

The ingredients are three: the **weave**, the **warp colour order** (the sequence of colours in which ends are drawn, e.g. "1 dark, 1 light" written **1 & 1**, or "4 dark, 4 light" = **4 & 4**), and the **weft colour order**. The **effect repeat** is the LCM of the three repeats in each direction: ends repeat = LCM(weave ends, warp order), picks repeat = LCM(weave picks, weft order). Colour-and-weave effects are at their best with two strongly contrasting colours and weaves of small, regular interlacing — plain and 2/2 twill are the classics.

2. First experiment — 1 & 1 colouring on plain weave

Warp order 1 dark, 1 light; weft order 1 dark, 1 light, both starting with dark; plain weave. Intuition expects a fine all-over mixture ("pepper-and-salt"). Compute instead, pick by pick (Topic 4 discipline: verify, don't guess). On pick 1 (dark weft): the ends up are the odd ends — the *dark* warp ends — and where the even (light) ends are down, the dark weft shows: **the whole pick appears dark**. On pick 2 (light weft): the ends up are the even, light ends; where the dark ends are down, the light weft shows: **the whole pick appears light**.

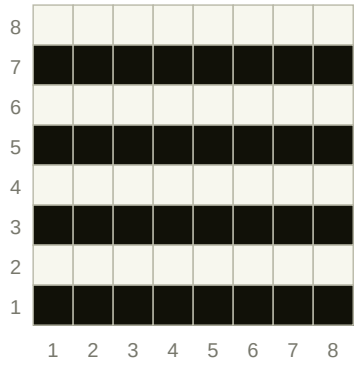


Figure 1 — Visible-colour grid for plain weave with 1 & 1 colour orders in both directions (black squares = dark yarn visible, white = light). The computed result is fine **horizontal (weft-way) stripes**, one pick high — the standard surprise of this topic, and the reason colour-and-weave work is always drawn out, never assumed.

Now shift the weft order by one pick (start the weft with light instead of dark): every conclusion reverses, and the same weave and warp order give **vertical (warp-way) stripes** one end wide. Two fabrics, identical records except for where the weft order starts — the cleanest demonstration in the course that in colour-and-weave work, *phase* is a design variable.

3. Second experiment — 2 & 2 colouring on plain weave

Double both orders: warp 2 dark, 2 light; weft 2 dark, 2 light, plain weave. Now each pick contains runs of both colours, and the effect breaks into a small-scale mixed check — dark blocks where dark warp stands over dark weft zones, light blocks opposite, and mixed hairline cells along the boundaries where the two systems' phases cross.

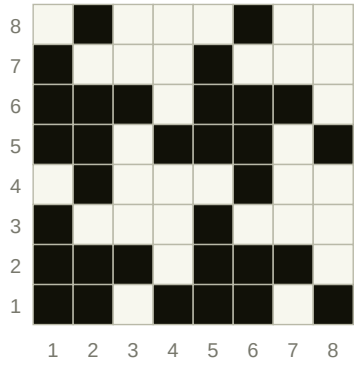


Figure 2 — Visible-colour grid, plain weave, 2 & 2 orders both ways (one effect repeat, 8 × 8). Neither stripes nor clean blocks: the characteristic small checkered mixture of the 2 & 2 plain colour-and-weave, from which shirting checks are built by scaling the orders up (4 & 4, 6 & 6…) — the mixture cells scale with them.

4. The classic — 4 & 4 colouring on 2/2 twill: dogtooth

Warp order 4 dark, 4 light; weft order 4 dark, 4 light; weave 2/2 twill (repeat 4). Effect repeat: ends LCM(4, 8) = 8, picks 8 — but the character only declares itself over **two repeats**, so Figure 3 tiles the

effect 2 × 2. In the dark-warp/dark-weft quadrant the square is solidly dark; in the light/light quadrant, solidly light; in the two cross quadrants the twill steps the boundary between the colours one end per pick, cutting the square corners into the jagged, abstract **dogtooth (houndstooth / pied-de-poule)** points that make the pattern unmistakable in suiting.

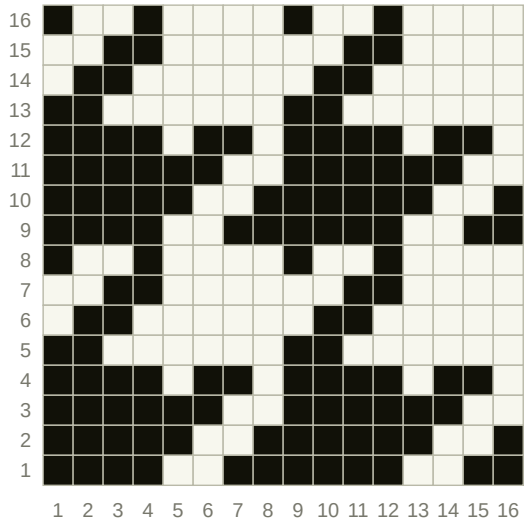


Figure 3 — Dogtooth: 2/2 twill with 4 & 4 colour orders, effect shown over 16 × 16 (two repeats each way). Solid dark and solid light blocks alternate with the stepped, broken-point blocks — the woven original of the printed houndstooth check.

EFFECT	WEAVE	ORDERS (WARP / WEFT)	APPEARANCE
Hairline stripes	Plain	1 & 1 / 1 & 1	One-thread stripes, direction set by the weft phase (Section 2)
Mixed check	Plain	2 & 2 / 2 & 2 and larger	Small blocks with mixed boundary cells (Figure 2)
Shepherd's check	2/2 twill	equal orders, e.g. 6 & 6 / 6 & 6	Clean square checks with twill texture, no broken points
Dogtooth (houndstooth)	2/2 twill	4 & 4 / 4 & 4	Blocks with jagged stepped points (Figure 3)
Pinstripe / pinhead	Plain or twill	orders with one contrasting thread per group, e.g. 11 & 1	Single-thread lines or dots on a solid ground
Bird's eye	pointed/diamond base	two-colour orders	Small diamond spots with a central dot — colour-and-weave on the Topic 3 diamond construction

5. Working rules

- **Draw the visible-colour grid** for at least one full effect repeat before committing yarn — every figure in this sheet is that drawing.
- **Contrast first:** the effects live on value contrast; two close colours collapse the whole mechanism into a mixture.
- **Phase is free:** shifting either order by one thread is a different design (Section 2) — record the starting thread of each order in the specification (TE-3206 does).
- **Balanced weaves show both colours;** strongly faced weaves (3/1, satin) suppress the minority system and mute the effect — one more reason plain and 2/2 dominate this field.
- The same computation, with the colour of dye lots and yarn counts as they will actually be woven, is how mills predict whether a buyer's check sample can be matched in bulk.

Glossary

Colour order — the repeating sequence of yarn colours in warp or weft (written 1 & 1, 4 & 4…). **Effect repeat** — LCM of weave repeat and colour orders in each direction. **Phase** — the starting thread of an order relative to the weave. **Dogtooth / houndstooth** — broken-point check from 2/2 twill with 4 & 4 orders. **Shepherd's check** — clean square check from equal larger orders on 2/2 twill.

Key points

• Visible colour = the colour of whichever thread is on top — a computation, not an impression.
• 1 & 1 on plain gives one-thread stripes (phase decides direction); 2 & 2 gives mixed checks; 4 & 4 on 2/2 twill gives dogtooth.
• Effect repeat = LCM of the three ingredient repeats, in each direction separately.

Practice

1. Compute the effect repeat (ends × picks) for a 2/1 twill with warp order 3 & 3 and weft order 2 & 2.
2. For Figure 1's fabric, the weft order is started one pick later. State the resulting effect and the one-word design variable responsible.
3. Why does replacing plain weave by 3/1 twill, orders unchanged (1 & 1 both ways), destroy the clean stripe of Figure 1?
4. A mill wants the dogtooth points twice as large. Which numbers change, and what happens to the effect repeat?

Answer key

1. Ends: LCM(3, 6) = 6. Picks: LCM(3, 4) = 12. Effect repeat **6 × 12**.
2. Vertical (warp-way) one-end stripes replace the horizontal ones; the variable is the **phase** of the weft order relative to the weave.
3. In 3/1 twill the warp is up over 3 of every 4 ends, so on most intersections the *warp* colour shows regardless of the pick: the face becomes warp-dominated, the weft colour appears only in the single down-lines, and the equal alternation that built the stripe is gone. (The warp's own 1 & 1 order then shows as fine warp-way striping instead.)
4. The colour orders double to 8 & 8 (both directions), weave unchanged. Effect repeat: ends LCM(4, 16) = 16, picks 16 — the pattern geometry is the same, at twice the thread count per block.
